Khan Academy

Mastery challenge

$$\sum_{n=1}^{\infty} \frac{(n+1)!}{n!}$$

True or false?

According to the n^{th} term test, the series diverges.

Choose 1 answer:

CORRECT (SELECTED)
True

INCORRECT
False

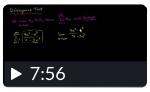
1 / 2

The nth term test is only conclusive when $\lim\limits_{n\to\infty} a_n \neq 0$.

2 / 2

Since $\lim\limits_{n\to\infty} \dfrac{(n+1)!}{n!} = \lim\limits_{n\to\infty} (n+1) \neq 0$, it is correct to say that the series diverges by the nth term test.

Related content



nth term divergence test

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